

ECO 340 DATA PROJECT:

Q4. The NLSY provides data on type of college/university degree obtained, as well as years of education. Do we see different wage returns to different degrees? How large are these different returns to field of study?

My interpretation of the question is as such and I have tried to answer these through my code.

MAIN QUESTION: Do hourly wages differ across highest degree obtained? Among people with postsecondary degrees, do wages also differ across fields of study?

I will be looking at answers to these questions:

1. Do wages differ across degree types?
2. Do those degree differences remain after controlling for the standard class controls?
3. Do those differences change when you add ASVAB?
4. Among degree holders, do wages differ across fields of study?
5. Does field still matter after controlling for degree type?

INTRODUCTION:

Returns to education in labour economics is an important study because it looks at the distribution of earnings and why similar workers earn different wages as seen in class. Return to education in terms of years of schooling is good but it neglects the important case that these differences may also differ across different levels of education/ degree such as Associates, Bachelors, Masters, PhD, etc (Buscha and Dickson 2023). Furthermore, even among works with the same post-secondary graduate degree the wages may differ because of the field of study they chose to pursue (Buscha and Dickson 2023). During class we looked at the fact that the estimated returns to education could be reflecting ability rather than just

education. So, we should be comparing it with and without an ability control like ASVAB. I am using the NLSY97 database. I will be using wage regressions using years of education and then replacing that with years of schooling with the highest degree obtained and then with the post-secondary degrees alone and then the different field of studies that those post-secondary degree holders hold.

DATA:

This data project uses the National Longitudinal Survey of Youth 1997 (NLSY97) as the data source. In particular I have been using the class `nlsy97_long` as the starting point dataset because it already contains all the main variable information needed such education, highest degree obtained, wages, hours worked, union status, sex, birth year, ASVAB, and more. My first steps were to load and clean the data, making the variables I was using as the benchmark and the controls present and cleaned. I needed to contrast the age variable which I did by converting the `birthyr` and the year to numeric form and then subtracting them. Before sample restrictions, age ranges from 11 to 44, with a mean age of about 27.49 and a median of 27. To construct a single years-of-education variable, I combine `highestgrade` for 1997, `highestd1` for 1998 to 2009, and `highestd2` for 2010 onward into one variable called `highestd`. Then, I turned the education values of 0 and 95 as invalid or missing values and recode them to NA. After that I sort the data by respondent and year and fill education downward within person so that missing education observations are replaced using later available information for the same respondent. The number of missing education observations falls a lot and the mean of `highestd` rises from 12.39 to 12.65. I cleaned the sample further by dropping the years 2012, 14, 16, 18, 20, 22, and 2024. I restricted the sample to respondents with labour income and positive employee hours in order to focus on valid wage observations. As done in the class and the tutorials, I use midpoint values based on the categorical income variable `workinc_est` to construct a usable work income measure since the exact work income is not directly

available. Hourly wage is calculated as annual work income divided by employee hours, and log hourly wage is constructed as the natural logarithm of hourly wage. In the cleaned wage sample, mean annual work income is about 27,849, mean hourly wage is about 21.54, and mean log hourly wage is about 2.276. Mean tenure in the cleaned wage sample is about 2.93 years, the female share is about 0.495, and mean ASVAB is about 49.23. Then I created the control variables needed for the regressions, age², age³, tenure, female dummy variable, ASVAB as an ability control. Because the data project focuses on more completed schooling outcomes, I also create a restricted sample of non-students aged 30 and above for the main wage regressions. In the age-restricted sample, potential labour-market experience is approximated as Age - Years of Education - 6, and its mean is about 15.45 years.

For the Field of study, I downloaded separate variables from the NLSY97 about college majors (Education -> school specific characteristic roster -> colleges attended -> By term -> Major). I first downloaded all the respondents from that section which came to around 400 values which didn't seem right, especially when I was doing the merging code. So, then I scrapped that plan and chose to download the variables from 2013 to 2023 and merge it to the nlsy cleaned and age restricted data set by the respondents ID. I clean the major data by converting all columns to numeric and recoding negative values and the code 999 as missing. 1,681 respondents have a non-missing coded major and the remaining 7,303 observations in the major file are missing a usable major code. I group these coded majors into broad field categories labelled Business, Education, Health, Humanities/Arts, Other, Social Science/Public Service, STEM, and Trades/Applied. After merging the field data onto the age-restricted wage sample, the final respondent-year sample contains 28,531 observations, of which 4,897 have a non-missing field group. I also construct a respondent-level degree variable using the maximum observed value of highested3 within respondent and recode it into Less than HS/GED, High School Diploma, Associate, Bachelor's, Master's, PhD, and

Professional degree categories. In the merged age-restricted sample, the largest education category is High School Diploma with 10,970 observations, followed by Bachelor's with 6,269 and Less than HS/GED with 4,765. For the field-of-study regressions, I further restrict the sample to respondents with a non-missing field group and a postsecondary degree, which leaves 2,895 observations and 828 unique respondents. Within this final field sample, the largest degree categories are Associate with 1,097 observations and Bachelor's with 1,205 observations, while PhD and Professional are much smaller categories with 13 and 50 observations respectively. I used figures to summarize the data, including education by age, mean log wage by degree, mean log wage by field of study, wage densities by field, and coefficient plots for field premiums. I estimate all main regressions using OLS and report clustered standard errors at the respondent level so that inference accounts for repeated observations on the same individual over time. The final model are from a simple years-of-education specification to degree-type regressions, and then to field-of-study regressions, and models that add both degree controls and ASVAB.

MODEL SETUP:

I will be using multiple estimates of OLS regressions in which the dependent variable is log hourly wage as shown in class. Our first regression is our benchmark regression and it uses log hourly wage on highested, age, age², age³, union, tenure, female and as year fixed as a reference point for the return to years of education so that I have something to compare to when I differentiate it with degrees and field of study. In the second regression I added ASVAB as an ability control to observe whether the estimated education premium becomes smaller once measure ability is held constant. In my third and fourth regression I start by replacing years of education with the degree categories to answer the question "Do wages differ across degree types?". Log hourly wage is regressed on degree type, age, age², age³, union, tenure, female and as year fixed. ASVAB is added to the degree model so that I can

test whether the estimated degree premiums shrink once measured ability is included. This ended up answering whether those differences remain after adding the standard controls, and whether they change when ASVAB is included. The fifth and sixth regression models focus on field of study and therefore restrict the sample to respondents with a non-missing field group and a postsecondary degree. Log hourly wage is regressed on field group, age, age², age³, union, tenure, female and as year fixed. The omitted field category is Social Science/Public Service, so each field coefficient is interpreted relative to that reference group. In reg 6, I add the degree type to the regression to answer the question that do different degree holders have wages differ across fields of study? The omitted degree category is Bachelor's, so the degree coefficients are interpreted relative to Bachelor's degree holders while the field coefficients remain relative to Social Science/Public Service. I add ASVAB to the field + degree model in order to test whether field premiums and degree premiums change once measured ability is controlled for. I report clustered standard errors at the respondent level for the main regression tables so that inference accounts for repeated observations on the same individual over time. This clustered standard error approach is appropriate because the dataset is a respondent-year panel rather than a simple cross section.

RESULTS:

```

Call:
lm(formula = lnwage ~ highested + age + agesq + agecu + union +
    tenure + female + as.factor(year), data = nlsy_agerestric)

Residuals:
    Min       1Q   Median       3Q      Max
-7.4209 -0.3559  0.0174  0.3790  7.1105

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)   -6.8688954   6.3371795   -1.084   0.27842
highested      0.1014897   0.0016544  61.345 < 0.0000000000000002 ***
age            0.6267746   0.5294040    1.184   0.23645
agesq        -0.0156978   0.0146220   -1.074   0.28302
agecu         0.0001308   0.0001337    0.978   0.32786
union         0.0254450   0.0135134    1.883   0.05972 .
tenure        0.0256116   0.0010335  24.781 < 0.0000000000000002 ***
female       -0.2354071   0.0099305  -23.705 < 0.0000000000000002 ***
as.factor(year)2011 -0.0647954   0.0348077   -1.862   0.06268 .
as.factor(year)2013 -0.0785909   0.0347411   -2.262   0.02369 *
as.factor(year)2015 -0.0623024   0.0391447   -1.592   0.11149
as.factor(year)2017  0.0844657   0.0438743    1.925   0.05422 .
as.factor(year)2019  0.0799781   0.0467745    1.710   0.08730 .
as.factor(year)2021  0.1539066   0.0497655    3.093   0.00199 **
as.factor(year)2023  0.4558711   0.0528176    8.631 < 0.0000000000000002 ***
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Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.7743 on 24995 degrees of freedom
(3521 observations deleted due to missingness)
Multiple R-squared:  0.2262,    Adjusted R-squared:  0.2258
F-statistic: 521.9 on 14 and 24995 DF,  p-value: < 0.0000000000000002

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	Basic	Robust SE	Clustered SE
highested	0.1015 *** (0.0017)	0.1015 *** (0.0017)	0.1015 *** (0.0026)
age	0.6268 (0.5294)	0.6268 (0.5256)	0.6268 (0.4790)
agesq	-0.0157 (0.0146)	-0.0157 (0.0145)	-0.0157 (0.0132)
agecu	0.0001 (0.0001)	0.0001 (0.0001)	0.0001 (0.0001)
union	0.0254 (0.0135)	0.0254 (0.0134)	0.0254 (0.0184)
tenure	0.0256 *** (0.0010)	0.0256 *** (0.0010)	0.0256 *** (0.0015)
female	-0.2354 *** (0.0099)	-0.2354 *** (0.0099)	-0.2354 *** (0.0151)
N	25010	25010	25010
R2	0.2262		

*** p < 0.001; ** p < 0.01; * p < 0.05.

In Regression 1: the coefficient on the years of education is 0.1015 and is significant at 0.1% level ($p < 0.001$). This means that each addition year of education is approximately a 10.2% higher hourly wage, holding age, tenure, union status, gender, and year fixed. The SE in

creases from 0.0017 to 0.0026 in the cluster SE model. Tenure at 0.0256 and females at -0.2354 are also significant because it shows that women earn approx. 23.5% less per hour than men. 0.2262 shows a 22.6% wage variation.

```
Call:
lm(formula = lnwage ~ highested + asvab + age + agesq + agecu +
    union + tenure + female + as.factor(year), data = nlsy_agerestric)

Residuals:
    Min       1Q   Median       3Q      Max
-7.4919 -0.3498  0.0169  0.3673  6.8993

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept) -3.19797748  6.82675123  -0.468  0.63947
highested    0.06650312  0.00219401  30.311 < 0.0000000000000002 ***
asvab        0.00625233  0.00022135   28.246 < 0.0000000000000002 ***
age          0.35541155  0.57032901    0.623  0.53318
agesq       -0.00879509  0.01575330   -0.558  0.57664
agecu        0.00007354  0.00014408    0.510  0.60974
union        0.06499327  0.01478430    4.396  0.0000111 ***
tenure       0.02436439  0.00111684   21.815 < 0.0000000000000002 ***
female      -0.23123678  0.01073328  -21.544 < 0.0000000000000002 ***
as.factor(year)2011 -0.06417966  0.03808536   -1.685  0.09197 .
as.factor(year)2013 -0.06725220  0.03792036   -1.774  0.07616 .
as.factor(year)2015 -0.02989843  0.04263048   -0.701  0.48310
as.factor(year)2017  0.11476795  0.04767447    2.407  0.01608 *
as.factor(year)2019  0.10340703  0.05078742    2.036  0.04176 *
as.factor(year)2021  0.17723515  0.05395628    3.285  0.00102 **
as.factor(year)2023  0.47420268  0.05718272    8.293 < 0.0000000000000002 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.7556 on 20564 degrees of freedom
(7951 observations deleted due to missingness)
Multiple R-squared:  0.2571,    Adjusted R-squared:  0.2565
F-statistic: 474.4 on 15 and 20564 DF,  p-value: < 0.00000000000000022
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	Education	+ASVAB
highested	0.1015 *** (0.0026)	0.0665 *** (0.0032)
asvab		0.0063 *** (0.0003)
age	0.6268 (0.4790)	0.3554 (0.5185)
agesq	-0.0157 (0.0132)	-0.0088 (0.0143)
agecu	0.0001 (0.0001)	0.0001 (0.0001)
union	0.0254 (0.0184)	0.0650 ** (0.0204)
tenure	0.0256 *** (0.0015)	0.0244 *** (0.0016)
female	-0.2354 *** (0.0151)	-0.2312 *** (0.0162)
nobs	25010	20580
logLik	-29082.2790	-23425.9920
AIC	58196.5600	46885.9800
BIC	58326.5900	47020.8300
nobs.1	25010.0000	20580.0000

*** p < 0.001; ** p < 0.01; * p < 0.05.

In regression 2, I have added ASVAB as an ability control. The education coefficient falls from 0.1015 to 0.0665 which represents ability bias. The drop means that more educated workers are also higher ability, and without ASVAB the education coefficient was both the return to education and wage premium. The ASVAB coefficient is 0.0063 with a $p < 0.001$, meaning that each additional percentile point of ability raises wages by 0.63% independent of education. The union premium rises from 0.0254 which is insignificant to 0.0650 with $p < 0.01$ after controlling for ability because higher ability workers are less likely to be union members. The sample also drops from 25010 to 20580 because of the missing ASVAB values.

	Degree (no ASVAB)	Degree + ASVAB
degree_typeAssociate	0.1836 *** (0.0255)	0.1400 *** (0.0277)
degree_typeBachelor's	0.4878 *** (0.0199)	0.3555 *** (0.0224)
degree_typeMaster's	0.5811 *** (0.0234)	0.4024 *** (0.0265)
degree_typePhD	0.6998 *** (0.0529)	0.4745 *** (0.0587)
degree_typeProfessional	1.1717 *** (0.0492)	0.8982 *** (0.0543)
degree_typeLess than HS / GED	-0.2494 *** (0.0234)	-0.1441 *** (0.0256)
asvab		0.0058 *** (0.0003)
age	0.6999 (0.4774)	0.3740 (0.5162)
agesq	-0.0176 (0.0132)	-0.0093 (0.0142)
agecu	0.0001 (0.0001)	0.0001 (0.0001)
union	0.0403 * (0.0184)	0.0739 *** (0.0203)
tenure	0.0249 *** (0.0015)	0.0243 *** (0.0016)
female	-0.2273 *** (0.0149)	-0.2272 *** (0.0159)
nobs	24991	20571
logLik	-28845.1930	-23242.4160
AIC	57732.3900	46528.8300
BIC	57903.0400	46703.3300
nobs.l	24991.0000	20571.0000

*** $p < 0.001$; ** $p < 0.01$; * $p < 0.05$.

In regression 3, I replaced the years of education with the degree type with High School Diploma as a reference category. As you can see all the degrees above HS diploma show a large, positive and highly significant premium. R^2 rises from 0.2262 in regression 1 to 0.2385

(please check the regression 3 model output) showing that the degree held by the workers explains wages better than the years of education. In Regression 4, we add ASVAB as an ability control to the degree model. We can see that while positive in reference to the HS diploma still fall and the amount fallen increases Associates to PhD. This shows that higher the degree held (minus the Professionals), the more of its apparent premium was reflecting ability rather than the degree type. The ASVAB being 0.0058 and highly significant in addition to everything else means that part of the wage premium associated with higher degrees is explained by measured ability, but the degree premiums remain large and highly significant, so degree type still matters even after ASVAB is controlled for.

	Field only	Field + Degree
field_groupSTEM	0.1478 (0.0803)	0.2080 ** (0.0780)
field_groupBusiness	0.2758 *** (0.0681)	0.2725 *** (0.0657)
field_groupHealth	0.2436 *** (0.0672)	0.2759 *** (0.0634)
field_groupHumanities/Arts	0.0780 (0.0937)	0.0403 (0.0903)
field_groupEducation	-0.0893 (0.0844)	-0.1842 * (0.0805)
field_groupTrades/Applied	0.1397 (0.0946)	0.2447 ** (0.0895)
field_groupOther	0.1034 (0.0837)	0.1481 (0.0836)
degree_typeAssociate		-0.2651 *** (0.0413)
degree_typeMaster's		0.1728 ** (0.0571)
degree_typePhD		0.4636 ** (0.1428)
degree_typeProfessional		0.6573 *** (0.1187)
union	0.1868 *** (0.0485)	0.1643 *** (0.0478)
tenure	0.0266 *** (0.0048)	0.0245 *** (0.0045)
female	-0.1703 *** (0.0433)	-0.1725 *** (0.0421)
nobs	2528	2528
logLik	-2742.4960	-2656.0580
AIC	5528.9920	5364.1160
BIC	5657.3660	5515.8310
nobs.1	2528.0000	2528.0000

*** p < 0.001; ** p < 0.01; * p < 0.05.

In regression 5, I have estimated the field of study premiums without degree controls with Social Science/ Public Service as a reference category. Business has a coefficient of 0.2758

and is highly statistically significant, Health has a coefficient of 0.2436 and is highly statistically significant, and STEM has a coefficient of 0.1478, which is positive and significant. Education is -0.0893 but not statistically significant, Humanities/Arts is 0.0780 and not statistically significant, Trades/Applied is 0.1397 and not statistically significant, and Other is only marginally significant. This shows that Business and Health have higher wages.

In regression 6, I added the degree types control with bachelor degree as reference. Business is strongly positive at 0.2725, Health is strongly positive at 0.2759, STEM rises to 0.2080, and Trades/Applied rises to 0.2447, while Education becomes more negative at -0.1842.

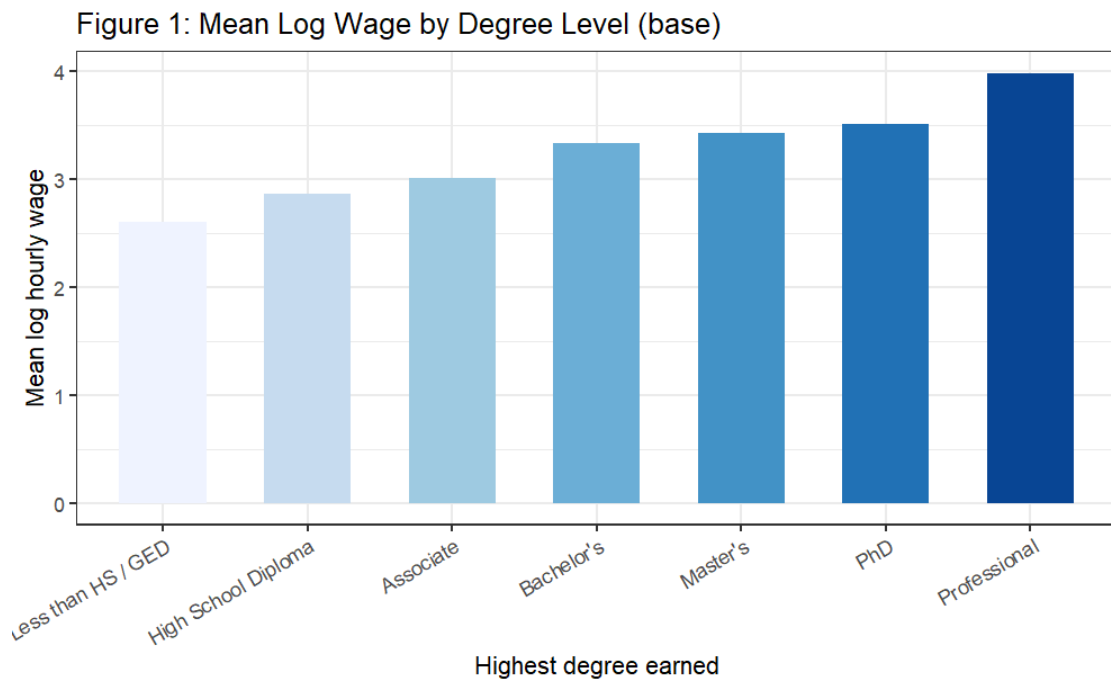
	Field + Degree (no ASVAB)	Field + Degree + ASVAB
field_groupSTEM	0.2080 ** (0.0780)	0.1694 (0.0875)
field_groupBusiness	0.2725 *** (0.0657)	0.2743 *** (0.0708)
field_groupHealth	0.2759 *** (0.0634)	0.2661 *** (0.0670)
field_groupHumanities/Arts	0.0403 (0.0903)	-0.0118 (0.1000)
field_groupEducation	-0.1842 * (0.0805)	-0.1957 * (0.0903)
field_groupTrades/Applied	0.2447 ** (0.0895)	0.2522 ** (0.0883)
field_groupOther	0.1481 (0.0836)	0.0821 (0.0872)
asvab		0.0036 *** (0.0009)
union	0.1643 *** (0.0478)	0.1800 *** (0.0538)
tenure	0.0245 *** (0.0045)	0.0266 *** (0.0052)
female	-0.1725 *** (0.0421)	-0.1830 *** (0.0471)
nobs	2528	2102
logLik	-2656.0580	-2206.0270
AIC	5364.1160	4466.0540
BIC	5515.8310	4618.6220
nobs.1	2528.0000	2102.0000

*** p < 0.001; ** p < 0.01; * p < 0.05.

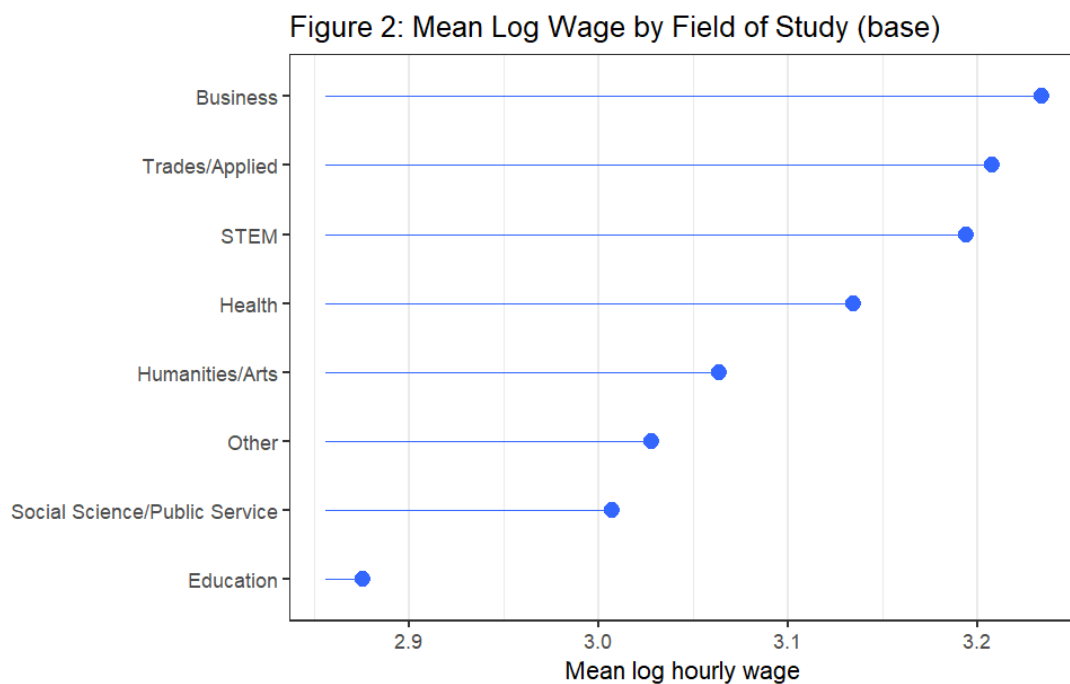
In regression 7, I added ASVAB as an ability control. The STEM falls from 0.2080 to 0.1694 and loses significance showing that it is still practical but not fully explained by ability.

Business as seen is almost unchanged, Health is still significant, education increases. ASVAB is at 0.0036 showing that cognitive ability is independently raising wages within the post-secondary education group sample.

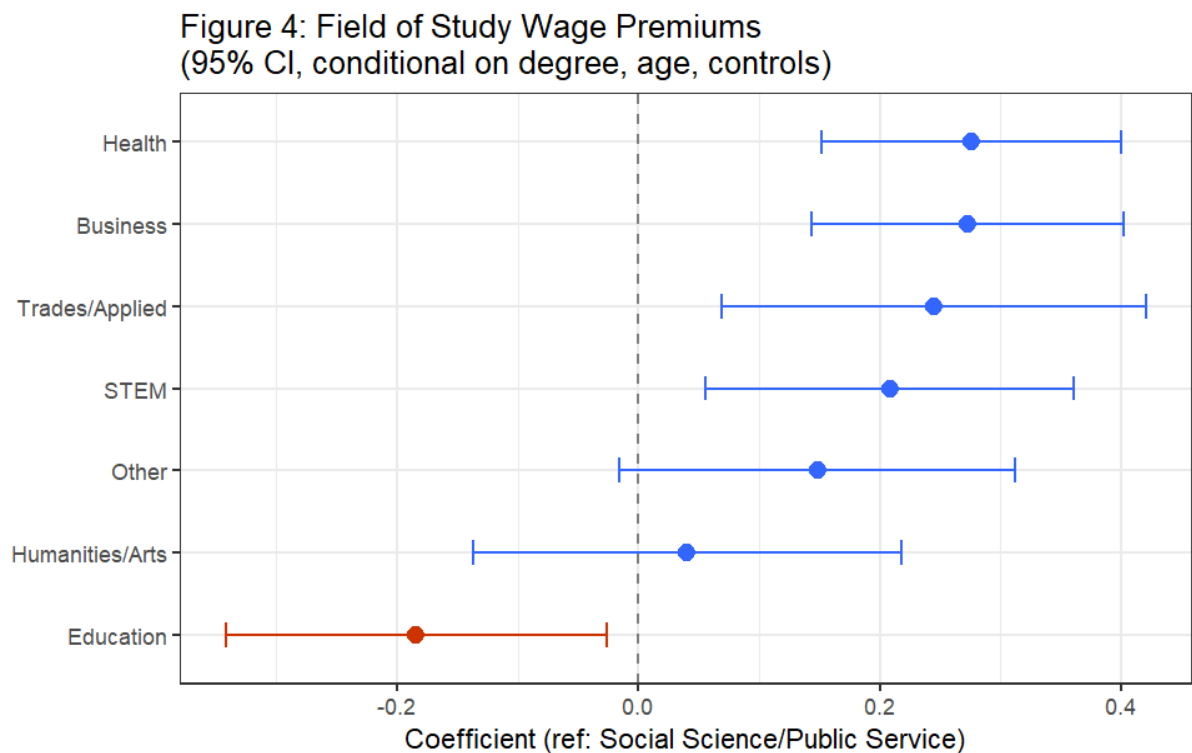
FIGURES:



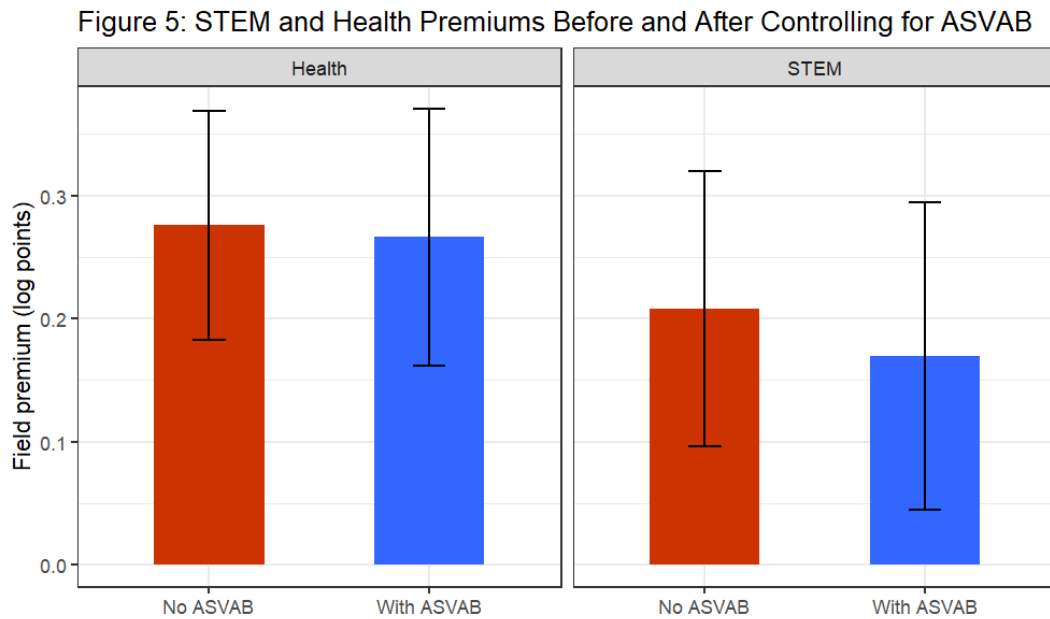
This is a mean log hourly wage for each degree type in the age restricted sample. mean log wage rises from Less than HS / GED to High School Diploma, to Associate, to Bachelor's, to Master's, to PhD, to Professional. This shows that workers with higher degrees earn higher average wages before any controls.



The figure ranks the fields of study by the average log hourly wage in the postsecondary field of study. Business has the highest mean log wage while Education has the lowest. This is the baseline difference before we add degree types and ASVAB as controls.



This is the coefficient plot with log wage regression on field of study and degree type, plus age, union, tenure, female, and year effects. The reference field is Social Science/Public Service. The dots are the estimated field coefficient and the line with it is the 95% CI. The dotted line represents that there is no difference from the reference field. This shows the regression result that some field differences remain even after controlling for degree type, which means the field effects are not only due to differences in level of degree obtained by the workers.



This compares the estimated field premiums for STEM and Health from the model without ASVAB and the model with ASVAB. Both fields still have positive estimated premiums after ASVAB is added, but the STEM premium becomes less robust than the Health premium.

LIMITATIONS:

1. The sample is also restricted to non-students aged 30 and older, which means the results are not generalizable to younger workers or current students. I didn't realise my mistake in the code until I had finished everything and was writing the data part of my report.
2. Even though ASVAB is added in several regression, measured ability is only one possible source of omitted-variable bias, and the models still do not control for all factors that may influence both education and wages.
3. The majors information is incomplete, since even after cleaning and regrouping, the number of respondents with usable major information is limited relative to the full wage sample. There is also my lack of understanding of the NLSY97 database which

makes me question if I have downloaded incomplete information. I also did not take in account the second major as a variable.

4. One of the limitations was when I was doing the the coalesce() step which selects the first non-missing major among the downloaded major fields in the order they appear in the code, so the assigned major_raw may not always represent the respondent's most recent or final completed major.

CONCLUSION:

Q1. Do wages differ across degree types?

- Yes, they do differ as we can see in reg 3 using the HS diploma as a reference point. All the degree categories have large and statistically significant coefficients besides Less than HS/ GED which was negative showing that workers with less than a high school credential earn significantly lower log hourly wages than workers with a high school diploma.

Q2. Do those degree differences remain after controlling for the standard class controls?

- Yes, differences in the type of degree remain after controlling for the standard class controls. Reg 3 includes age, age², age³, union status, tenure, female, and year fixed effects, and the degree coefficients remain large and highly statistically significant.

Q3. Do those differences change when you add ASVAB?

- The difference in degrees become smaller after ASVAB is added as an ability control. The fact that the degree coefficients fall after ASVAB is added shows that part of the wage premium associated with higher degrees reflects measured ability, but the

coefficients remain large and highly significant, so ability does not explain the entire degree premium.

Q4. Among degree holders, do wages differ across fields of study?

- Yes, wages do differ amongst degree holders of different fields of study as seen in regression 5. The different coefficients discussed in the results section show that field of study does matter for wages, but the strength of the premium differs across fields.

Q5. Does field still matter after controlling for degree type?

- Yes, it does as seen in regression 6. The differences are not simply reflecting the fact that some workers have higher degrees than others, because the field effects remain even after degree type is held constant as seen by the results.

Generative AI Statement:

A. Report writing:

1. I used AI to help me improve my wording, clarity and remove informal tones in lines such as [I needed to contrast the age variable ... information for the same respondent], [I restricted the sample to ... valid wage observations.], etc.
 - a. Prompt used: Can you make this more clear since it is too muddled?
2. I used AI to understand what the cluster SE meant from my regression and where I should be adding that.
 - a. Prompt used: why are clustered SE imp for regressions?
3. I used AI to clean up my writing while writing the results to make them to the point while keeping all of my writing and observations.
 - a. Prompt used: There is too much data in this paragraph and my point is getting lost. Can you just tell me without changing my words what I should do to make it flow better.
4. There were some regressions and some outputs where I had to ask what the results meant but wrote them myself.

B. Coding:

1. I used AI to identify and fix bugs in the R code, such as a column naming mismatch in the field of study data file, an incorrect ordering of the coalesce function that was selecting the most recent rather than earliest major and missing variable constructions.
 - a. Prompt used: Check my R code for ECO 340 project. Tell me what is wrong and what is missing
2. I did not know how to merge the majors variables I downloaded so I asked AI to tell me how to do that.

- a. This is the majors dat file and I have loaded it on the code but I don't know how to merge it with the nlsy. Can you tell me how to do it?
3. I asked to check my entire code for any big errors I had done and to see if I missed anything.
 - a. Prompt used: Check my R code for ECO 340 project. Tell me what is wrong and what is missing
4. I asked AI to help me with my figures codes because they were not loading and they looked wrong. It also ended up giving me what additional figures I should be putting in.
 - a. This is my entire code. The figures are not pretty and are not looking correct. What have I done wrong?
5. While submitting I was having issues with the rendering because the pdf version looked really bad and the codes were running out of the page so I asked it for help
 - a. its is not a good pdf the code is going outside the page and everything. What should I do ?

Bibliography

Buscha, Franz , and Matt Dickson. 2023. “Returns to Education: Individuals.” *Handbook of Labor, Human Resources and Population Economics* , January, 1–39.

https://doi.org/10.1007/978-3-319-57365-6_377-1.